

Abhirama M

Electronics and Communication Engineer

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Professional Summary

Dedicated Electronics and Communication graduate with a strong foundation in digital electronics, firmware development (C/C++, Python), and 32-bit microcontrollers. Eager to contribute to innovative embedded product development.

Proficient in bare-metal programming, RTOS, peripheral interfacing (I2C, SPI, UART), and wireless communication (LoRa, Bluetooth, WiFi). Skilled in hardware troubleshooting and Linux system administration.

Education

- 2023–2026 **B.E. in Electronics and Communication**, Kalpataru Institute of Technology, Tiptur, CGPA: 7.56
- 2020–2023 **Diploma in Electronics and Communication**, Sanjay Memorial Polytechnic, Sagara, CGPA: 9.26
- 2010–2020 **High School**, Pragathi Composite School, Sagara, 58.56%

Technical Skills

- Hardware** ESP32, RP Pico, ARM Cortex-M (LPC1768), KiCad, PCB Layout, Sensors, Logic Analyzer, DMM.
- Embedded** C, C++, Python, MicroPython, Firmware Development, RTOS, I2C, SPI, UART, EEPROM, LoRa, RF, Bluetooth, WiFi.
- Software** Linux Administration, PlatformIO, Keil IDE, GCC, ncurses, Thonny IDE, Git, Bash.
- Languages** English, Kannada

Experience

- Oct 2025–Feb 2026 **Embedded Systems Intern**, Cranes Varsity Pvt Ltd
- Completed hands-on embedded systems training including microcontroller programming, peripheral interfacing (UART, ADC, GPIO), debugging, and application development using ARM Cortex-M (LPC1768), C, and Keil IDE.

Projects

- Mar 2026–Apr 2026 **PumpPilot (IoT Control)**, ESP32
- ESP32-based automatic pump control system with timer-based operation, Bluetooth app support, LoRa long-distance control, and memory-based setting storage.
- Tech/Tools:* C/C++, PlatformIO, Relays, LoRa, Bluetooth.
- Feb 2026 **Intelligent POS System**, LPC1768
- Microcontroller-based POS billing system integrated with Python GUI software for bill data management through UART communication.
- Tech/Tools:* Embedded C, Keil, Python, Tkinter, 4x4 Keypad, 16x2 LCD.
- Dec 2025 **Traffic Simulator**, Linux/Terminal
- Terminal-based traffic simulation project developed in C focused on signal control and vehicle flow simulation.
- Tech/Tools:* C Programming, ncurses library, GCC Compiler.
- Jun 2025–Oct 2025 **Universal Electronic Meter**, RP2040 (Pico)
- Compact multi-purpose electronic measuring device for monitoring electrical parameters, reducing the need for multiple bulky instruments.
- Tech/Tools:* MicroPython, Thonny IDE, Voltage/Current Sensor Circuits.